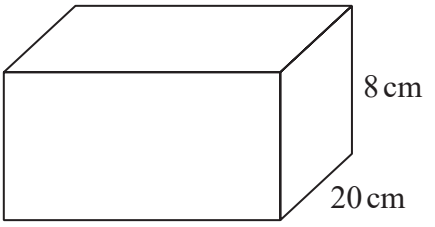
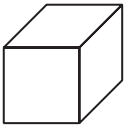
13 A box is in the shape of a cuboid 32 cm by 20 cm by 8 cm.

Box



Brick



Sam has 75 bricks.
Each brick is a cube of side length 4 cm.

Can Sam fit all 75 bricks into the box?
You must show how you get your answer.

Along the box: 32 ÷ 4 = 8 bricks

1 mark
Each dimension is a multiple of 4 cm.

Across the box: 20 ÷ 4 = 5 bricks

Layers: 8 ÷ 4 = 2 bricks

Capacity = 8 × 5 × 2 = 80 bricks

1 mark
Count all rows and layers.

Yes. 75 ≤ 80, so all 75 fit, with 5 spaces left.

1 mark
The dimension counts show a physical arrangement that fits.

(Total for Question 13 is 3 marks)

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